5. $\underset{\text{acid 1}}{H_2O} + \underset{\text{base 1}}{H_2O} \rightleftharpoons \underset{\text{acid 2}}{H_3O^+} + \underset{\text{base 2}}{OH^-}$

$\underset{\text{acid 1}}{HF} + \underset{\text{base 1}}{H_2O} \rightleftharpoons \underset{\text{acid 2}}{H_3O^+} + \underset{\text{base 2}}{F^-}$

$\therefore$ Conjugate bases are OH^- and F^- respectively

i.e.,[Option (c)]

6. Basic buffer is the solution which has weak base and its salt

$$\underset{200\,ml}{NH_4OH} + \underset{100\,ml}{HCl} \longrightarrow \underset{Salt}{NH_4Cl} + H_2O + \underset{(100\,ml\ weak\ base)}{NH_4OH}$$

i.e.,[Option (c)]

7. $BF_3 \rightarrow$ elctron deficient $\rightarrow$ Lewis acid

$PF_3 \rightarrow$ electron rich $\rightarrow$ lewis base

$CF_4 \rightarrow$ neutral $\rightarrow$ neither lewis acid nor base

$SiF_4^- \rightarrow$ neutral $\rightarrow$ neither lewis acid nor base

[option (b)]

8. $BF_3 \rightarrow$ elctron deficient $\rightarrow$ Lewis acid

$PF_3 \rightarrow$ electron rich $\rightarrow$ lewis base

$CO \rightarrow$ having lone pair of electron $\rightarrow$ lewis base

$F^- \rightarrow$ unshared pair of electron $\rightarrow$ lewis base

[option (a)]

9. $HCOONa + H\cdot OH \rightleftharpoons \underset{\text{strong base}}{NaOH} + \underset{\text{weak acid}}{H\text{-}COOH}$

Basic in nature.

$$C_6H_5NH_3Cl^- + H\cdot OH \rightleftharpoons \underset{acidic}{H_3O^+} + C_6H_5\text{-}NH_2 + Cl^-$$

$$KCN + H - OH \rightleftharpoons \underset{\text{strong base}}{KOH} + \underset{\text{weak acid}}{HCN}$$

basic

[option (b)] basic, acidic, basic is correct.

10. $C_5H_5N + H\text{-}OH \rightleftharpoons C_5H_5\overset{+}{N}H + OH^-$

$$\frac{\alpha^2 C}{1-\alpha} = K_b$$

$$\alpha^2 C \approx K_b$$

$$\alpha = \sqrt{\frac{K_b}{C}} = \sqrt{\frac{1.7\times10^{-9}}{0.1}}$$

$$= \sqrt{1.7}\times10^{-4}$$

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